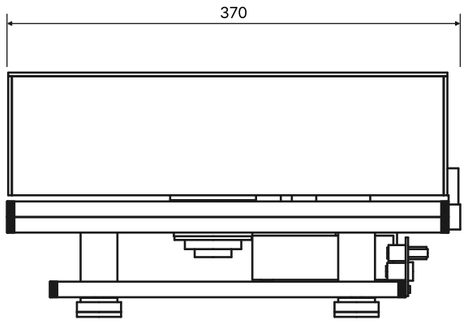
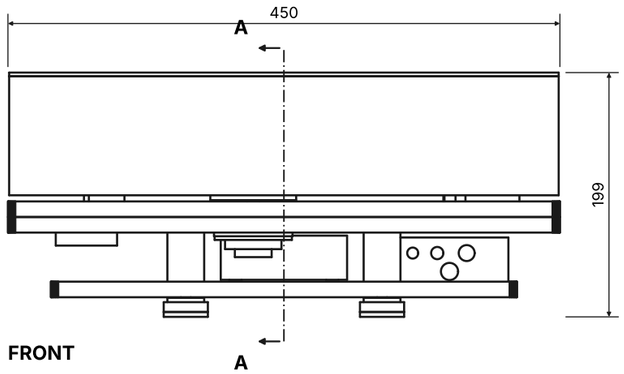
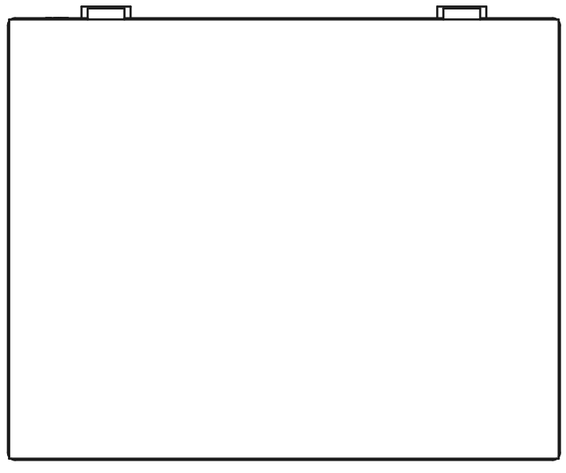
Scale 1:5. First angle: the view from the right is drawn to the left of the front, the view from above under it. A-A: the section on sheet 2.



FROM THE RIGHT



FRONT



FROM ABOVE

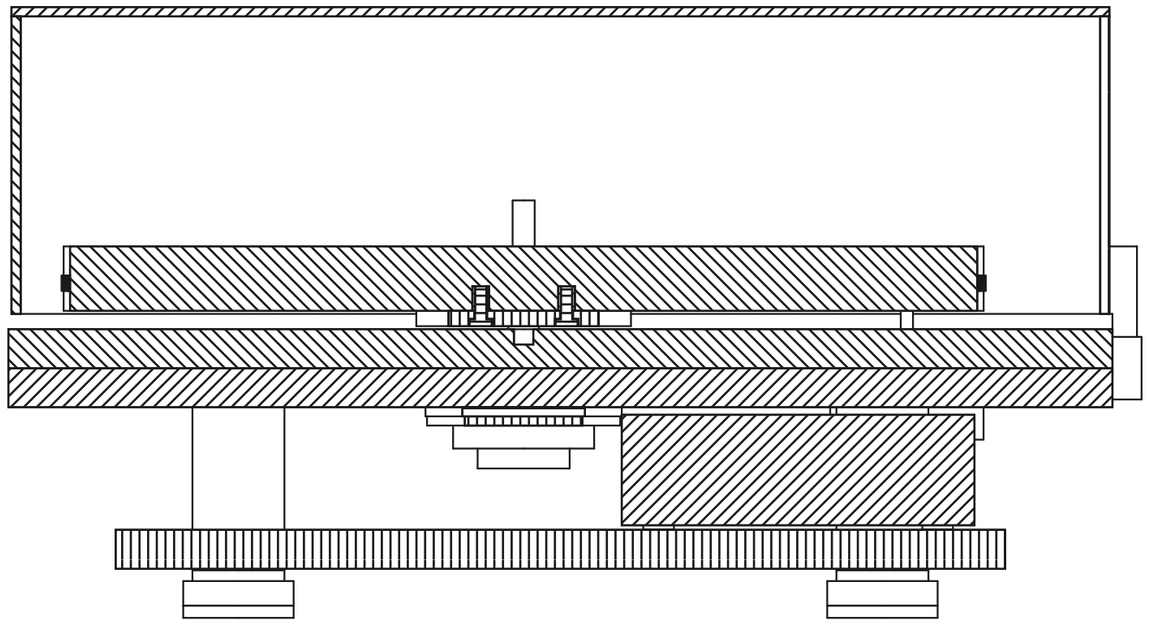
General arrangement

Mantle: Strata Float v3 (proposal A): Belt-drive record player, a tinted deck floating over a smoked base

mm | issue 2026-11-10 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:2. Section A-A (sheet 1): the product cut at x 235.0 and seen from the right; cut faces hatched, a different angle for each part.



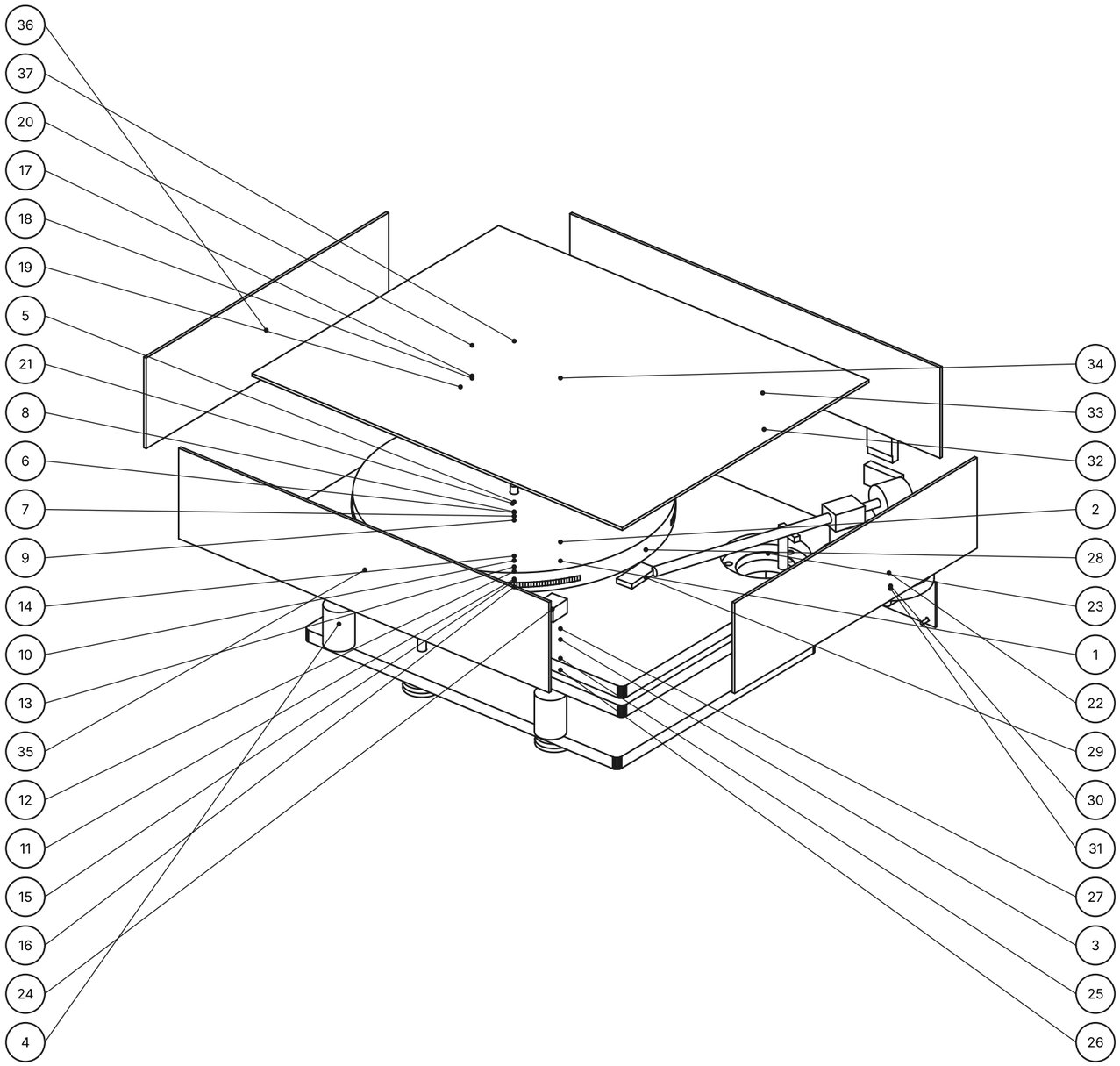
Section A-A

Mantle: Strata Float v3 (proposal A): Belt-drive record player, a tinted deck floating over a smoked base

mm | issue 2026-11-10 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5, parts drawn apart. Balloons are the parts list's numbers (last sheet).



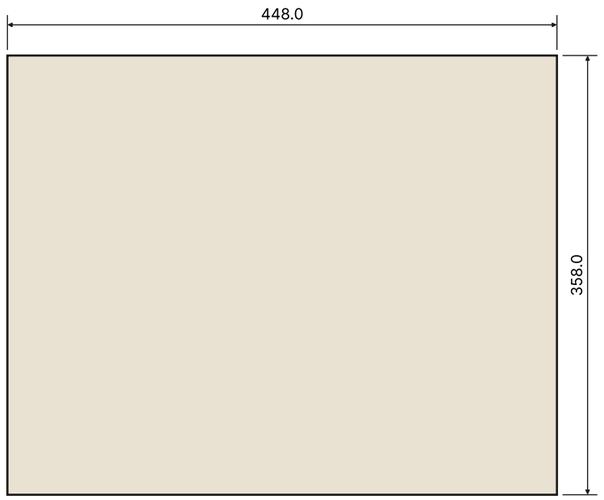
Exploded view

Mantle: Strata Float v3 (proposal A): Belt-drive record player, a tinted deck floating over a smoked base

mm | issue 2026-11-10 | first angle

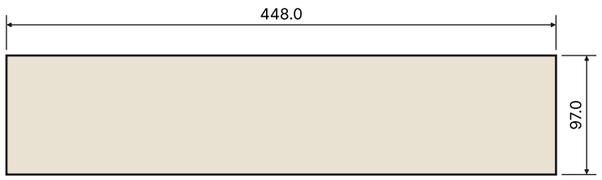
Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. Panels cut to their outline only (DXF files in dxf/); sizes as seen from the outer face.



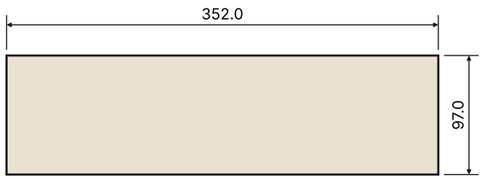
lid-top

3 mm trucast-tint, 1 for the set



lid-front, lid-back

3 mm trucast-tint, 2 for the set



lid-left, lid-right

3 mm trucast-tint, 2 for the set

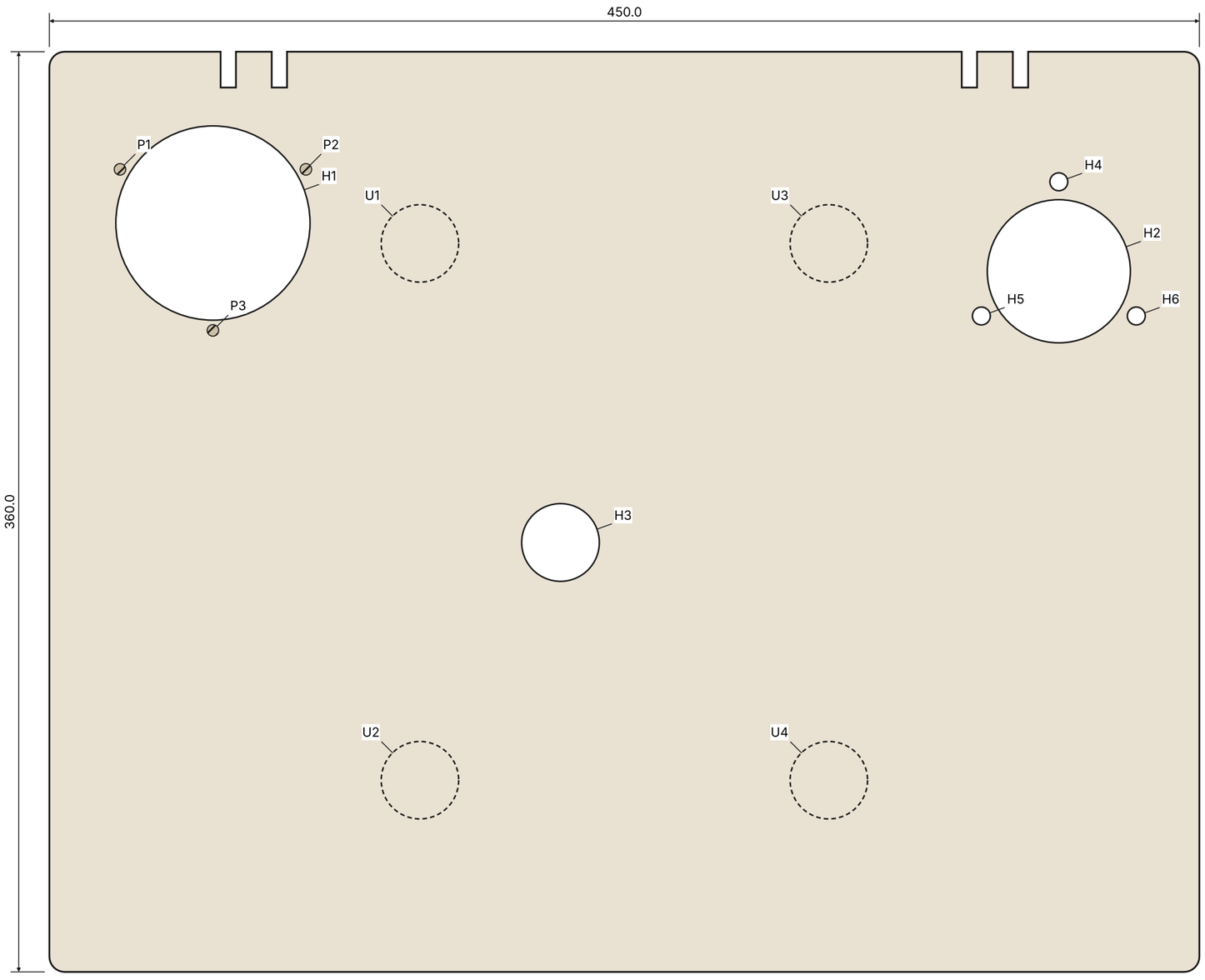
Sheet parts: plain panels

Mantle: Strata Float v3 (proposal A): Belt-drive record player, a tinted deck floating over a smoked base

mm | issue 2026-11-10 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:2. As the router sees it from the outer face; DXF: dxf/plinth-L1.dxf. Marks listed at the right.



mark	size	x	y	depth
P1	Ø4.6	27.6	314.0	1.3 from face
P2	Ø4.6	100.4	314.0	1.3 from face
P3	Ø4.6	64.0	251.0	1.3 from face
H1	Ø76.0	64.0	293.0	through
H2	Ø56.0	395.0	274.1	through
H3	Ø30.4	200.0	168.0	through
H4	Ø7.0	395.0	309.1	through
H5	Ø7.0	364.7	256.6	through
H6	Ø7.0	425.3	256.6	through
U1	Ø30.3	145.0	285.0	2 from inside
U2	Ø30.3	145.0	75.0	2 from inside
U3	Ø30.3	305.0	285.0	2 from inside
U4	Ø30.3	305.0	75.0	2 from inside

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

plinth-L1: 12.7 mm trucast-tint, 1 for the set, seen from its outer face

inner-face pockets: dxf/plinth-L1-underside.dxf (seen from that face)

not in the cut file: the opening at 70, 353 rounded over from the outer face (1 faces, 0 to 0 deg off the sheet)
and: the opening at 90, 353 rounded over from the outer face (1 faces, 0 to 0 deg off the sheet)
and: the opening at 360, 353 rounded over from the outer face (1 faces, 0 to 0 deg off the sheet)

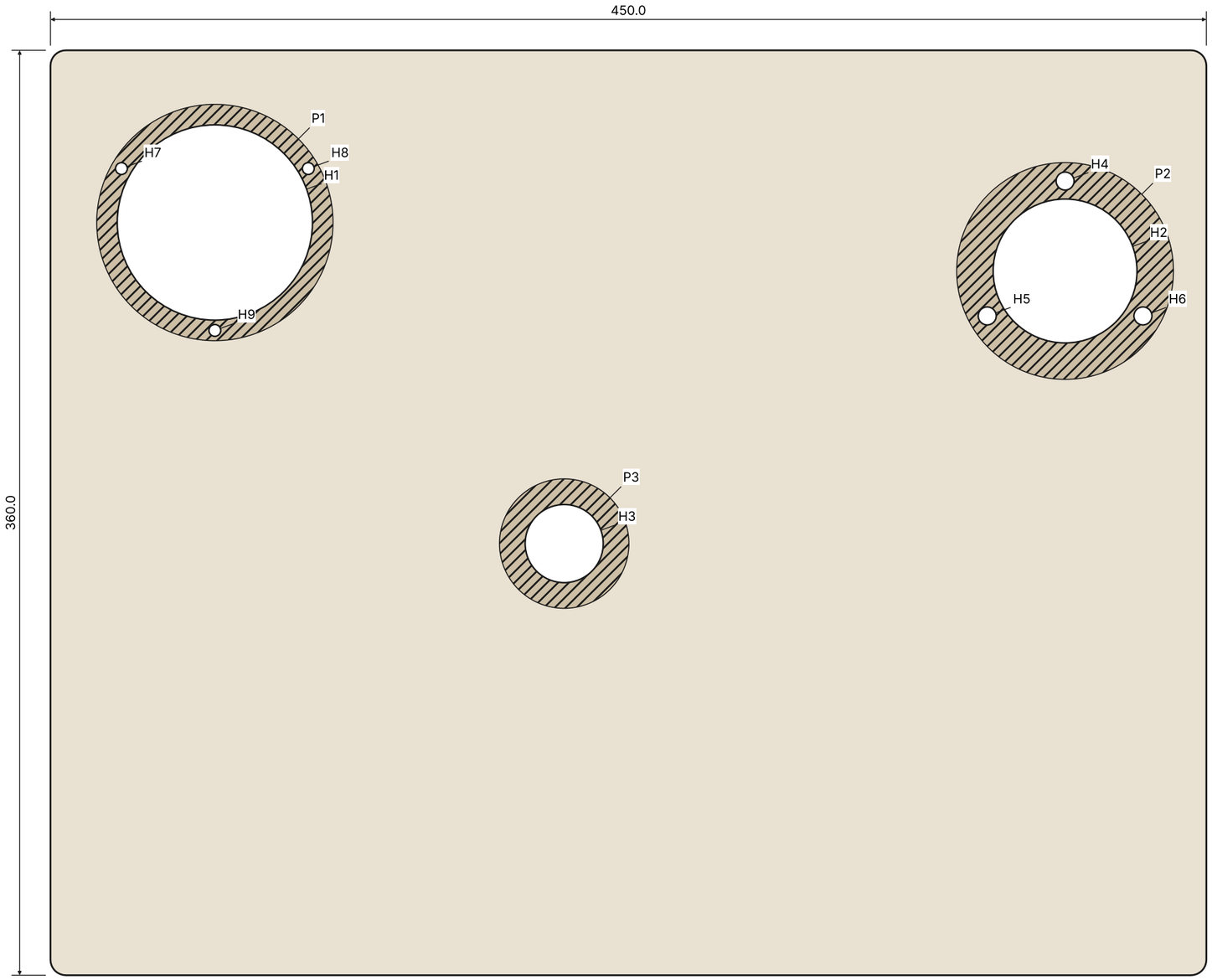
Sheet part: plinth-L1

Mantle: Strata Float v3 (proposal A): Belt-drive record player, a tinted deck floating over a smoked base

mm | issue 2026-11-10 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:2. As the router sees it from the outer face; DXF: dxf/plinth-L2.dxf. Marks listed at the right.



mark	size	x	y	depth
P1	Ø92.0	64.0	293.0	6 from face
P2	Ø84.4	395.0	274.1	12 from face
P3	Ø50.4	200.0	168.0	5 from face
H1	Ø76.0	64.0	293.0	through
H2	Ø56.0	395.0	274.1	through
H3	Ø30.4	200.0	168.0	through
H4	Ø7.0	395.0	309.1	through
H5	Ø7.0	364.7	256.6	through
H6	Ø7.0	425.3	256.6	through
H7	Ø4.6	27.6	314.0	through
H8	Ø4.6	100.4	314.0	through
H9	Ø4.6	64.0	251.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

plinth-L2: 12.7 mm trucast-tint, 1 for the set, seen from its outer face

not in the cut file: the opening at 70, 353 rounded over from the outer face (1 faces, 0 to 0 deg off the sheet)
and: the opening at 90, 353 rounded over from the outer face (1 faces, 0 to 0 deg off the sheet)
and: the opening at 360, 353 rounded over from the outer face (1 faces, 0 to 0 deg off the sheet)

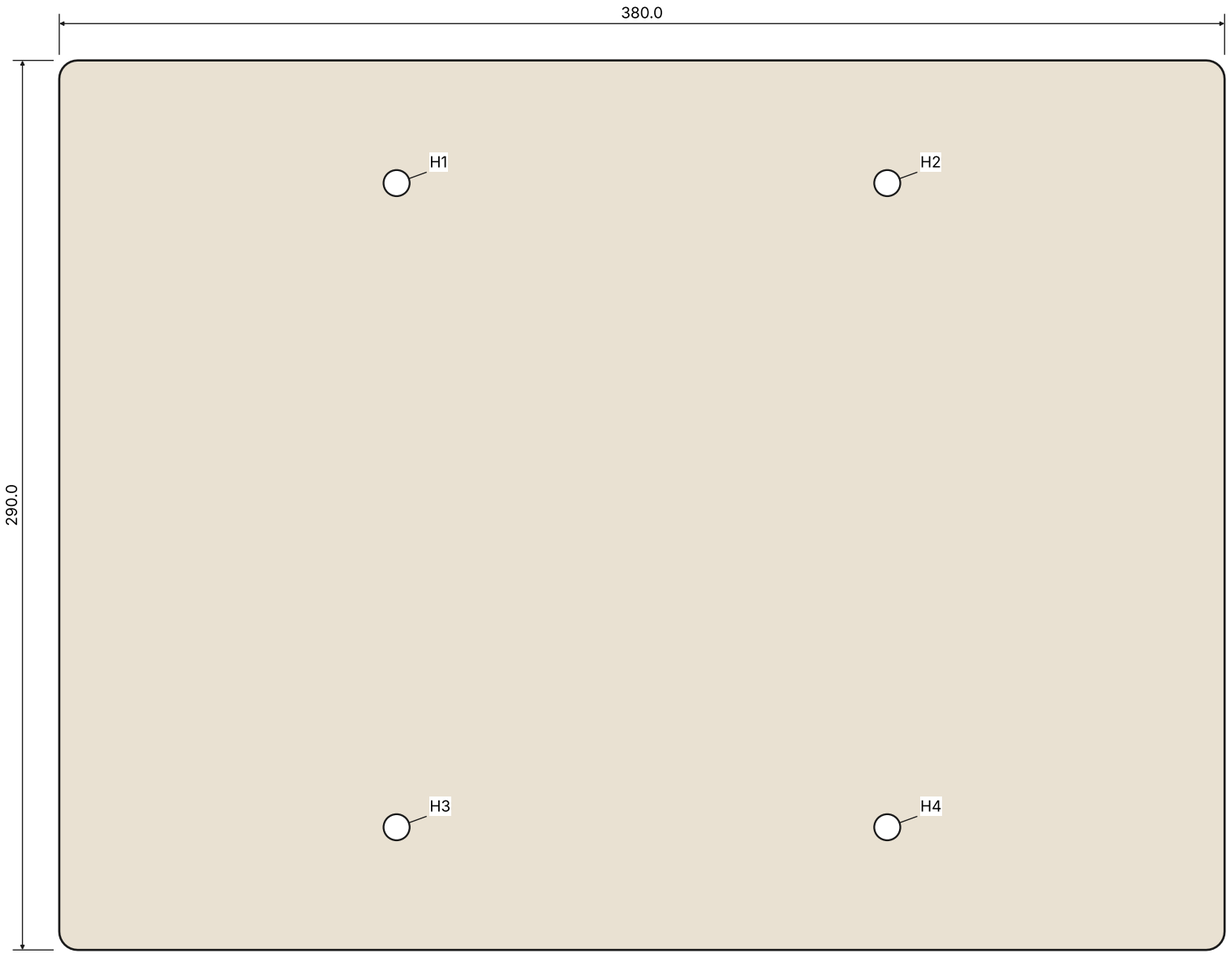
Sheet part: plinth-L2

Mantle: Strata Float v3 (proposal A): Belt-drive record player, a tinted deck floating over a smoked base

mm | issue 2026-11-10 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:2. As the router sees it from the outer face; DXF: dxf/base-L1.dxf. Marks listed at the right.



base-L1: 12.7 mm trucast-tint, 1 for the set, seen from its outer face

mark	size	x	y	depth
H1	Ø8.5	110.0	250.0	through
H2	Ø8.5	270.0	250.0	through
H3	Ø8.5	110.0	40.0	through
H4	Ø8.5	270.0	40.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

Sheet part: base-L1

Mantle: Strata Float v3 (proposal A): Belt-drive record player, a tinted deck floating over a smoked base

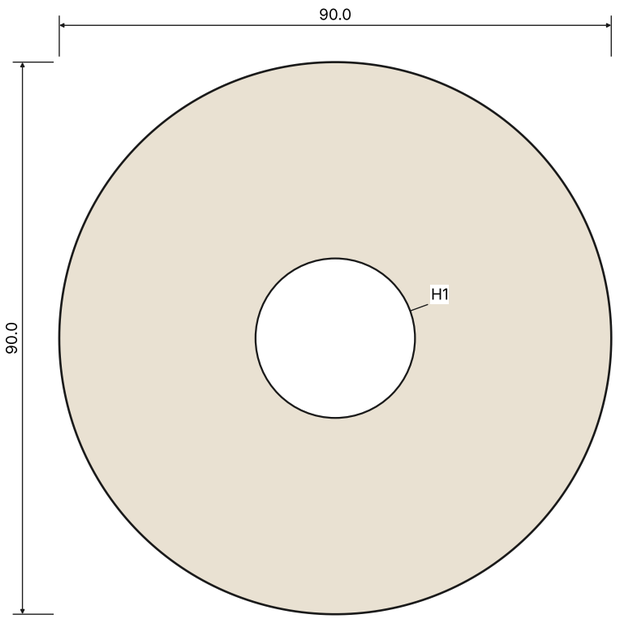
mm | issue 2026-11-10 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:1. As the router sees it from the outer face; DXF: dxf/motor-plate.dxf. Marks listed at the right.

mark	size	x	y	depth
H1	Ø26.0	45.0	45.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



motor-plate: 3 mm aluminium, 1 for the set, seen from its outer face

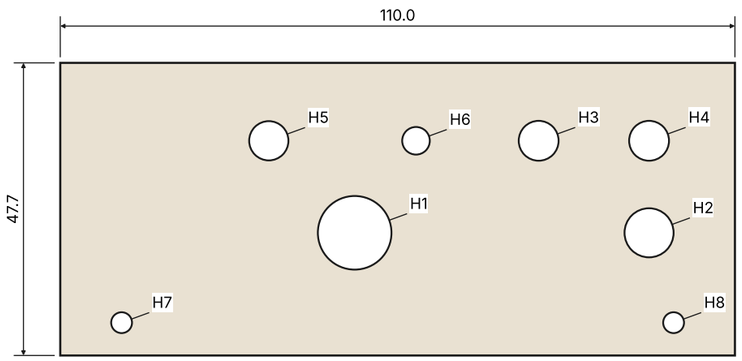
Sheet part: motor-plate

Mantle: Strata Float v3 (proposal A): Belt-drive record player, a tinted deck floating over a smoked base

mm | issue 2026-11-10 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:1. As the router sees it from the outer face; DXF: dxf/rear-plate.dxf. Marks listed at the right.



rear-plate: 2 mm brass-sheet, 1 for the set, seen from its outer face

mark	size	x	y	depth
H1	Ø12.0	48.0	20.0	through
H2	Ø8.0	96.0	20.0	through
H3	Ø6.5	78.0	35.0	through
H4	Ø6.5	96.0	35.0	through
H5	Ø6.4	34.0	35.0	through
H6	Ø4.5	58.0	35.0	through
H7	Ø3.4	10.0	5.4	through
H8	Ø3.4	100.0	5.4	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

Sheet part: rear-plate

Mantle: Strata Float v3 (proposal A): Belt-drive record player, a tinted deck floating over a smoked base

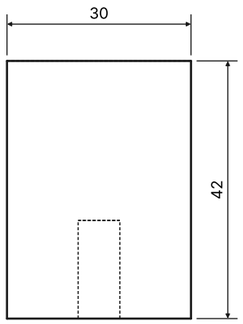
mm | issue 2026-11-10 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

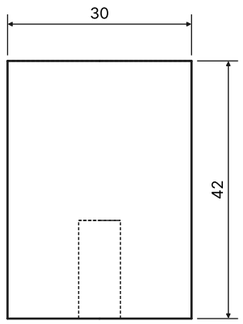
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

4. post-1, post-2, post-3, post-4: lathe, stainless; 29 cm3 each; 4 for the set; 1:1

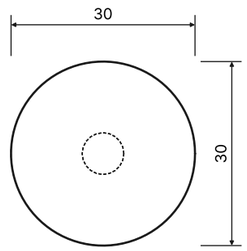
solid 303 stainless 30 × 42, turned, ends faced; finished satin black (the owner, 2026-11-10: black posts, the lamps' satin black #18181A): hot black oxide for stainless, satin, oiled (stainless kept for the load and the bond; black oxide over black chrome for its satin, not mirror, finish); tapped M8 × 16 from below; bonded 2 mm up into a seat under the deck with a structural methacrylate (Plexus MA310 kind, which bonds acrylic to stainless); the foot's stud holds it to the base from below: nothing shows



FRONT



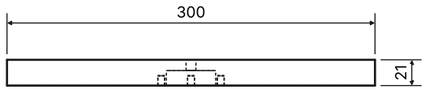
FROM THE RIGHT



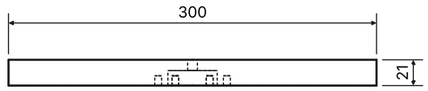
FROM ABOVE

5. platter: lathe, cast-acrylic; 1467 cm3 each; 1 for the set; 1:5

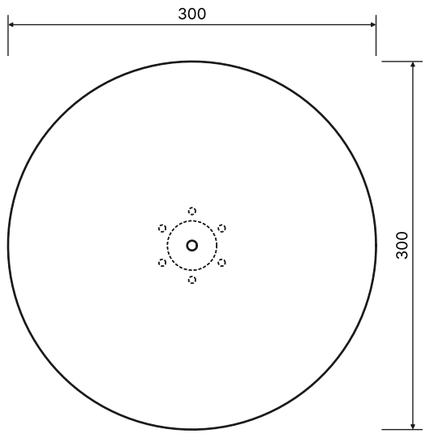
tinted 3 / clear 12.7 / tinted 5.6 cemented, annealed, turned to 300 × 21 as Outcrop's; six blind 5.6 holes from below for bonded brass M4 inserts, the record side unbroken



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

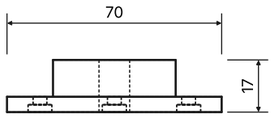
Mantle: Strata Float v3 (proposal A): Belt-drive record player, a tinted deck floating over a smoked base

mm | issue 2026-11-10 | first angle

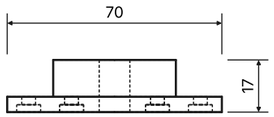
Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

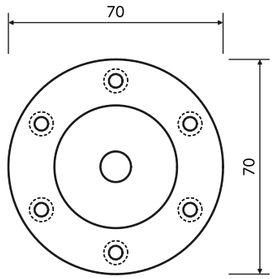
6. platter-hub: lathe, brass-leadfree; 32 cm3 each; 1 for the set; 1:2



FRONT



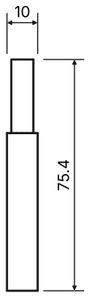
FROM THE RIGHT



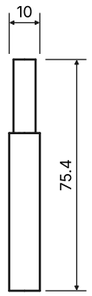
FROM ABOVE

9. spindle: lathe, stainless-440c; 5 cm3 each; 1 for the set; 1:2

as Outcrop's: 440C, 58 HRC, ground 10 g5; a vent flat 1.0 x 0.3 along the journals



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Mantle: Strata Float v3 (proposal A): Belt-drive record player, a tinted deck floating over a smoked base

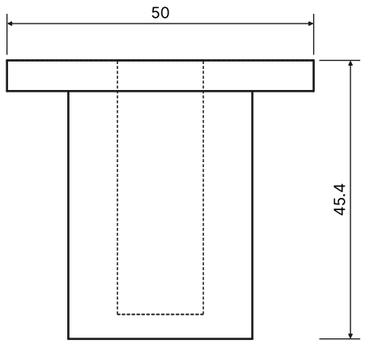
mm | issue 2026-11-10 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

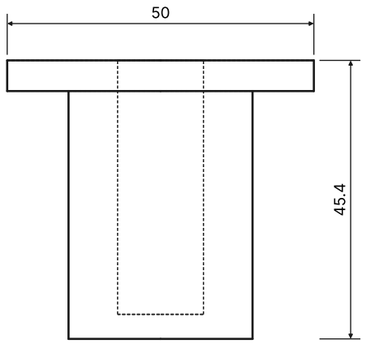
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

10. bearing-housing: lathe, brass-leadfree; 32 cm3 each; 1 for the set; 1:1

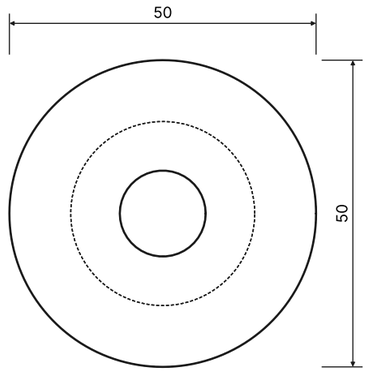
as Outcrop's: M30 × 1.5, clamped by its nut on a disc spring and a brass washer under the plinth (about 1.5 kN), hanging 20 into the float; its visible parts (housing, nut, washer) black (2026-11-10, to match the posts): brass, black nickel plated, satin; 1.0 ml of oil



FRONT

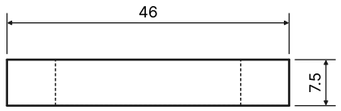


FROM THE RIGHT

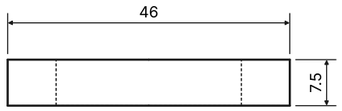


FROM ABOVE

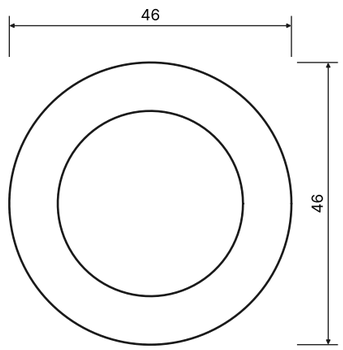
11. bearing-nut: lathe, brass-leadfree; 7 cm3 each; 1 for the set; 1:1



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

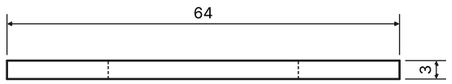
Mantle: Strata Float v3 (proposal A): Belt-drive record player, a tinted deck floating over a smoked base

mm | issue 2026-11-10 | first angle

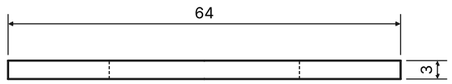
Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

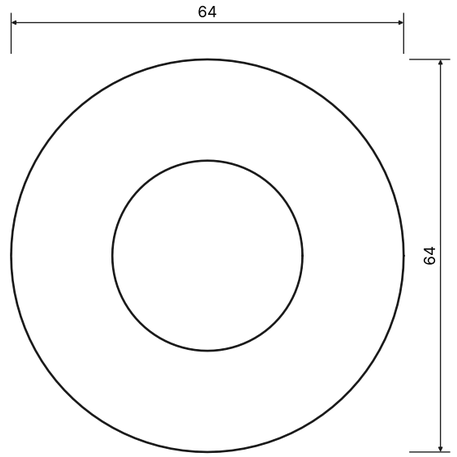
13. bearing-washer: lathe, brass-leadfree; 7 cm3 each; 1 for the set; 1:1



FRONT



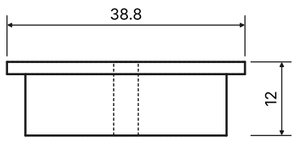
FROM THE RIGHT



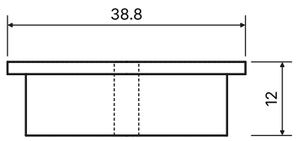
FROM ABOVE

20. pulley: lathe, aluminium; 11 cm3 each; 1 for the set; 1:1

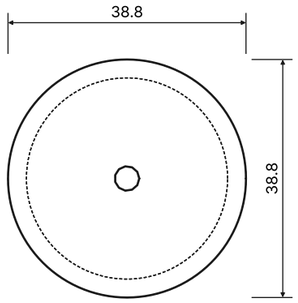
belt face 32.80 at 300 rpm: PROPOSAL, not turned until the bench test (STOCK-AND-AUDIT.md 3.1)



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

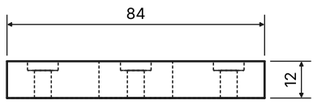
Mantle: Strata Float v3 (proposal A): Belt-drive record player, a tinted deck floating over a smoked base

mm | issue 2026-11-10 | first angle

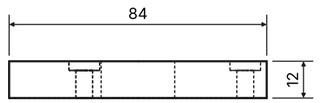
Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

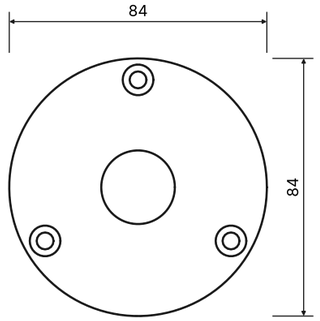
22. arm-puck: lathe, aluminium; 60 cm3 each; 1 for the set; 1:2



FRONT



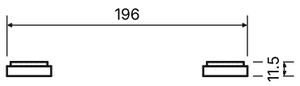
FROM THE RIGHT



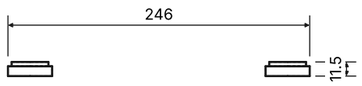
FROM ABOVE

25. feet: lathe, stainless; 42 cm3 each; 1 for the set; 1:5

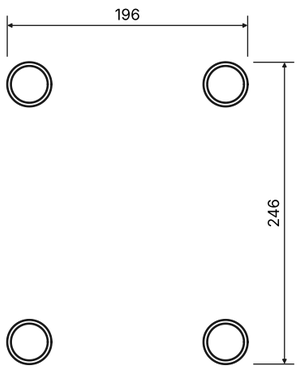
four, directly under the posts (one grid), turned 303 stainless, 36 × 8 with a 30 neck, on M8 studs through the base into the posts, levelled and locked; Sorbothane under. They locate and level; no isolation is claimed



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Mantle: Strata Float v3 (proposal A): Belt-drive record player, a tinted deck floating over a smoked base

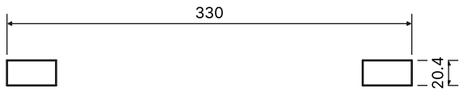
mm | issue 2026-11-10 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

32. hinge-blocks: cnc-mill, cast-acrylic; 16 cm3 each; 1 for the set; 1:5

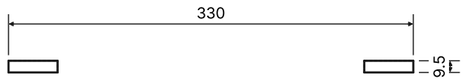
two 40 × 9.5 blocks of the color, each with the hinge's slot 3.2 × 12 × 18 deep, on two M4 screws into inserts in the plinth's back edge (horizontal, in shear)



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Mantle: Strata Float v3 (proposal A): Belt-drive record player, a tinted deck floating over a smoked base

mm | issue 2026-11-10 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

For 1; prices are budgets (bom.md says where each comes from).

no.	item	make	qty	what	size	each USD
1	plinth-L1	cut	1	12.7 mm Cell-cast acrylic sheet, stock tint in the line codes (TruCast 2423 Ember, 2092 Poison, 2287 Haze, 2069 Ice, 2074 Smoke), 4 × 8 ft	450 × 360	56.58
2	plinth-L2	cut	1	12.7 mm Cell-cast acrylic sheet, stock tint in the line codes (TruCast 2423 Ember, 2092 Poison, 2287 Haze, 2069 Ice, 2074 Smoke), 4 × 8 ft	450 × 360	56.58
3	base-L1	cut	1	12.7 mm Cell-cast acrylic sheet, stock tint in the line codes (TruCast 2423 Ember, 2092 Poison, 2287 Haze, 2069 Ice, 2074 Smoke), 4 × 8 ft	380 × 290	38.49
4	post-1, post-2, post-3, post-4	lathe	4	Stainless steel 304	35 × 35 × 46	16.69
5	platter	lathe	1	Cast acrylic tube, tinted through	310 × 310 × 22	169.57
6	platter-hub	lathe	1	Brass C69300 (Eco Brass, lead-free) or C27450	80 × 80 × 27	70.92
7	platter-screws	buy	1	Six M4 × 10 stainless socket button screws (ISO 7380), up through the hub into the platter's inserts		6.00
8	platter-inserts	buy	1	any (E-Z Lok / Tappex kind) Six M4 brass threaded inserts for bonding into plastics, 5.6 OD x 8, epoxy-bonded into the platter from below (no thread-locker near acrylic)		9.00
9	spindle	lathe	1	Stainless steel 440C, hardened 58 HRC and ground	20 × 20 × 85	127.05
10	bearing-housing	lathe	1	Brass C69300 (Eco Brass, lead-free) or C27450	60 × 60 × 55	74.92
11	bearing-nut	lathe	1	Brass C69300 (Eco Brass, lead-free) or C27450	56 × 56 × 18	53.23
12	bearing-spring	buy	1	Schnorr / Mubea / Christianson kind One DIN 2093 group 2 series C disc spring 63 x 31 × 1.8 (51CrV4), under the bearing nut; load and rate to confirm in the maker's table		6.00
13	bearing-washer	lathe	1	Brass C69300 (Eco Brass, lead-free) or C27450	74 × 74 × 13	55.68
14	bushes	buy	1	any lathe shop Two solid bronze plain bushes (C932 / SAE 660), 10 × 14 × 12, pressed into the housing and honed or line-reamed to 10 H6 (8-12 um running clearance on a 10 g5 spindle)		40.00
15	ball	buy	1	Tungsten carbide ball, 6 mm, grade 10: the spindle's thrust point, seated in a cup at the spindle's foot		5.00
16	thrust-plate	buy	1	Thrust plate 13.8 × 3, hardened 52100 steel (60 HRC) lapped flat, at the foot of the housing's bore in the oil bath, under the carbide ball (a sapphire window of the same size is the alternative)		20.00
17	motor-plate	cut	1	3 mm Aluminium 5052 sheet	90 × 90	1.62
18	motor	buy	1	Rega 24 V low-vibration synchronous motor upgrade kit (motor, its board and mounting), for the Neo PSU		265.00
19	motor-screws	buy	1	Three M3 × 12 stainless button screws and three M3 brass inserts (4.6 OD), the motor plate hard to the pocket's shoulder as Rega fits its motor		6.00
20	pulley	lathe	1	Aluminium 6061	49 × 49 × 22	46.31
21	belt	buy	1	Flat turntable belt, ground rubber or polyurethane, 5 wide x 0.6, cut to the measured path (about 945 long for the 994 path, 5 % stretch); an O-ring test belt first		30.00
22	arm-puck	lathe	1	Aluminium 6061	94 × 94 × 22	49.86

Parts list (1 of 3)

Mantle: Strata Float v3 (proposal A): Belt-drive record player, a tinted deck floating over a smoked base

mm | issue 2026-11-10 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

For 1; prices are budgets (bom.md says where each comes from).

no.	item	make	qty	what	size	each USD
23	arm	buy	1	Rega RB330 tonearm (9 in, Rega geometry: 222 mm from the platter's center to the arm's hole, effective length 236.5, overhang 14.5, offset 21.55 deg), with its own lead and cueing lift		675.00
24	cartridge	buy	1	Ortofon 2M Black moving-magnet cartridge (nude Shibata stylus), black body, 1/2 in mount		750.00
25	feet	lathe	1	Stainless steel 304	40 × 40 × 72	48.46
26	foot-pads	buy	1	Isolate It (Sorbothane Inc.) Three Sorbothane discs 40 × 4, 50 duro, bonded under the brass feet: about 45 N each, 36 kPa		20.00
27	foot-studs	buy	1	Three M8 × 30 stainless studs, three M8 brass lock nuts (thin), three M8 × 12 brass threaded inserts (for acrylic, epoxy-bonded)		18.00
28	phono-stage	buy	1	Pro-Ject Phono Box S2, MM/MC phono stage in an aluminum case, 103 × 115 × 36 (one listing 103 × 36 × 115 mm, 0.57 kg): RIAA within 0.4 dB 20 Hz-20 kHz, MM S/N 94 dBA at 40/43 dB gain, THD under 0.01 %, subsonic filter 20 Hz 12 dB/oct, MM 47k		269.00
29	phono-pads	buy	1	Four 10 mm felt pads under the phono stage		2.00
30	rear-plate	cut	1	2 mm Lead-free brass sheet (C26000 / C27450), brushed	110 × 48	2.10
31	rear-hardware	buy	1	Neutrik / Switchcraft / C&K / CUI kind For Strata v3 with no light (2026-11-10): two gold insulated panel RCA jacks (1/4 in hole); a 4 mm earth binding post; a 4PDT signal toggle with gold contacts (C&K 7411 kind); an insulated 2.1 mm DC		62.00
32	hinge-blocks	cnc-mill	1	Cast acrylic tube, tinted through	45 × 15 × 58	46.57
33	hinges	buy	1	Pro-Ject Replacement dust cover hinge for Debut series (rp-611), two: friction hinges that hold the lid open at any angle, a spring blade clamped to the lid's back wall, the foot in a slot (here a pocket in an acrylic block on the plinth's		34.00
34	lid-top	cut	1	3 mm Cell-cast acrylic sheet, stock tint in the line codes (TruCast 2423 Ember, 2092 Poison, 2287 Haze, 2069 Ice, 2074 Smoke), 4 × 8 ft	448 × 358	13.23
35	lid-front, lid-back	cut	2	3 mm Cell-cast acrylic sheet, stock tint in the line codes (TruCast 2423 Ember, 2092 Poison, 2287 Haze, 2069 Ice, 2074 Smoke), 4 × 8 ft	448 × 97	3.59
36	lid-left, lid-right	cut	2	3 mm Cell-cast acrylic sheet, stock tint in the line codes (TruCast 2423 Ember, 2092 Poison, 2287 Haze, 2069 Ice, 2074 Smoke), 4 × 8 ft	352 × 97	2.82
37	mark	buy	1	Laser etching into the cast acrylic, frosted, about 0.1 deep, from a vector file: EARF 44 wide, the etch filled with the color, deeper		15.00
38	inserts	buy	1	any (E-Z Lok / Tappex kind) Brass threaded inserts for plastics, epoxy-bonded: 3 M5 (the puck), 4 M3 (the back plate); with M5 × 12 countersunk and M3 × 6 button screws		14.00
39	wiring	buy	1	Shielded interconnects inside (the arm's lead to the S2, the S2 to the plate), a star earth at the plate's post (arm, bearing housing via an M3 ring tag, S2), two clear cord clips under the deck, a cord grip for the Neo's lead		25.00
40	neo-psu	buy	1	Rega Neo PSU MK2: DSP-generated drive for Rega's 24 V motor, 33/45 by a button, fine speed trim (0.01 rpm steps) on the back; US 60 Hz 115-120 V version		445.00
41	bearing-oil	buy	1	Turntable bearing oil, 8 ml (a light synthetic oil, ISO VG 32 or as the bush maker says)		15.00

Parts list (2 of 3)

Mantle: Strata Float v3 (proposal A): Belt-drive record player, a tinted deck floating over a smoked base

mm | issue 2026-11-10 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

For 1; prices are budgets (bom.md says where each comes from).

no.	item	make	qty	what	size	each USD
42	cement	buy	1	Evonik (Röhm) ACRIFIX 2R 0190, two-component polymerising cement for cast acrylic (cures through any tint, unlike the one-part light-curing 1R 0192); Weld-On 40 or 42 is the US equivalent		45.00

Parts list (3 of 3)

Mantle: Strata Float v3 (proposal A): Belt-drive record player, a tinted deck floating over a smoked base

mm | issue 2026-11-10 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Notes

1. The deck (as built since v2): two 12.7 sheets of the flavor's stock tint, cemented flush with ACRIFIX 2R 0190 in a vacuum press (25.4, one bond line at mid-edge), annealed, fly-cut, the outline routed as one; the outer edge bead-blasted satin, the faces glassy. 450 × 360, corners R4. (v1's tinted-ring / clear-inlay strata, 24.3, are gone.)
2. The base: one 12.7 sheet of stock Smoke 2074, 380 × 290, corners R6, 35 in from the deck all round; the deck floats 40 over it.
3. The float: four solid 303 stainless posts bonded into 2 mm seats under the plinth, held to the base by the feet's M8 studs: rigid. The four feet (stainless on Sorbothane, levelled and locked) locate and level it; no isolation is claimed. For isolation, a platform at 3-5 Hz.
4. No light (2026-11-10, the owner): no strip, channel, groove, adapter, switch or 24 V inlet. The deck's underside bead-blasted satin as its edge (the day win from the light proposals). The phono stage runs from its own listed adapter.
5. The rear plate: 2 mm brass on the base's back edge, under the deck's overhang; line out, earth post (the star earth), 4PDT stage / bypass, the S2's 18 V inlet and the Neo's; the rating engraved. Wiring and earth: strata_rear.py.
6. The motor: Rega 24 V on its plate, screwed hard to a shoulder in a pocket through the plinth as Rega fits it, below it in the float. The pulley waits for the motor's speed measured on the US Neo (300 rpm expected at 60 Hz).
7. The bearing, platter, arm puck and phono stage as Outcrop's (mantle_outcrop.py): a disc spring under the bearing's nut, 1.0 ml of oil, a vented 440C spindle; the platter screwed from below into bonded inserts; a 4PDT bypass.
8. Before the first sample: with the arm in hand, check its counterweight clears the lid's back wall.
9. The lid: 3 mm 2064, 1.1 kg, on Pro-Ject's friction hinges.

Notes

Mantle: Strata Float v3 (proposal A): Belt-drive record player, a tinted deck floating over a smoked base

mm | issue 2026-11-10 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15